

Introduction to computing

Case study: Comma separated values

columns are comma separated
it is a table

Roll no	Name	A1	A2	A3	A4
14	Talha	61	64	52	45

new column
new line show
new row-

writing CSV files:

```
' , '.join(['a', 'b', 'c'])
```

```
'a,b,c'
```

Exception handling :-

It is not in

C but in python

it exist.

user ki tamam galat ko handle
karna exception handling kehlata
ha-

- v # remember this
name error → filename
- f = open(" ~ ", 'mode')
IO error
- int('23')
23
- int('abc')
value error

Other example: database connectivity, internet connectivity,
bad values by users.

gracefull degradation:

باہر سے ملنے والے کام کی

$\text{div}(4, 0) \rightarrow$ how we handle this

like
if
else

```
try:  
    print "Div 4 by 2", div(4, 2)  
    print "    "    , 0, div(4, 0)  
except: zero division error  
    print "do not ask to div by 0"  
    print "done"
```

trycatch block


```
def calculator(op, n1, n2):  
    if op == 'div':  
        return div(n1, n2)  
    elif op == 'mul':  
        return mul(n1, n2)
```

```
print calculator(div, 4, 0)
```

Exception hierarchy



in hierarchy main thing exception

Raising exception

```
def sqrt(x, guess=1.0)  
    if x < 0: valueerror "not find sqrt  
    return none          of -ve num"
```


Classes

many real world problems can be modeled through classes.

- Any "object" in the world will belong to a class.
- Every class will have some properties and some functions.

information $\rightarrow$ classes
function

```
class Animal:
```

```
    legs = False
```

```
    name = ''
```

```
    def sound(self):
```

```
        return 'silence'
```

```
    def move(self):
```

```
        return True
```

`a = Animal()` # this is creating an instance (object)

type a → it denotes instance of class

instance

a.legs

False